# ML Model Evaluation & Explainability Report

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## 1. Dataset & Problem Detection Summary

| | | | |
|---|---|---|---|
| **Dataset File:** | house_prices.csv | **Total Rows:** | 2000 |
| **Target Column:** | Sale_Price | **Total Columns:** | 9 |
| **Problem Type:** | Regression | **Test Split:** | 20% |

## 2. Preprocessing & Validation Protocol

| | | | |
|---|---|---|---|
| **Numerical Imputer** | Median | **Categorical Encoding** | Onehot |
| **Categorical Imputer** | Most_frequent | **Feature Scaling** | Standard |
| **Random State** | 42 | **CV Protocol** | 5 Folds |

## 3. Model Performance Comparison

| Rank | Model | MAE | MSE | RMSE | $R^2$ | Adjusted $R^2$ | Train Time (s) |
|---|---|---|---|---|---|---|---|
| 1 | Linear Regression | 20678.54 | 643499045.11 | 25367.28 | 0.9815 | 0.9809 | 4.11 |
| 2 | Ridge Regression | 20720.47 | 645316444.97 | 25403.08 | 0.9814 | 0.9809 | 2.52 |
| 3 | Decision Tree Regressor | 35175.00 | 2050668350.00 | 45284.31 | 0.9410 | 0.9393 | 0.16 |
| 4 | Random Forest Regressor | 25732.74 | 1036394108.52 | 32193.08 | 0.9702 | 0.9693 | 1.37 |
| 5 | Gradient Boosting Regressor | 22567.80 | 780642583.89 | 27939.98 | 0.9775 | 0.9769 | 1.03 |
| 6 | XGBoost Regressor | 24641.18 | 943488297.80 | 30716.25 | 0.9728 | 0.9721 | 0.78 |

## 4. Recommended Champion Model

**Recommended Model: Linear Regression**
**Criterion: Highest $R^2$ (0.9815).**
**Rationale: Superior performance and fold consistency on held-out validation data.**

## 5. Global Feature Importance (SHAP)

| Feature | Mean Absolute SHAP Value | Relative Impact |
|---|---|---|
| Square_Feet | 161609.5767 | 100.0% |
| Neighborhood_Suburbs | 15584.5158 | 9.6% |
| Year_Built | 15186.4940 | 9.4% |
| Bathrooms | 11894.2884 | 7.4% |

| | | |
|---|---|---|
| Neighborhood_Uptown | 11813.6800 | 7.3% |
| Garage_Cars | 11525.8794 | 7.1% |
| Condition_Rating | 10744.3041 | 6.6% |
| Neighborhood_Downtown | 9794.8287 | 6.1% |